



ADITYA UNIVERSITY

Database Management System

Cornerstone Project: Library Management System

Department of Artificial Intelligence and Machine Learning

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Presentation outline

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Abstract

The Library Management System is a software application designed to automate and simplify library activities. It manages books, members, issue and return transactions, and library records. The system reduces manual paperwork and human errors while improving efficiency. It provides quick access to book availability and member information. Overall, it offers an organized and reliable way to manage library resources.



M A N A G E S

Books

Members

Issue & return

Library records

Introduction



A Library Management System is developed to manage the day-to-day activities of a library.



It stores information about books, members, librarians, and transactions.



The system allows librarians to add, update, search, issue, and return books.



It helps maintain accurate records and reduces the time required for manual operations.



The system improves the overall quality and efficiency of library services.

Problem statement



Traditional library management involves maintaining many records manually.



Searching for books and checking availability can be time-consuming.



Manual issue and return records may lead to errors or misplaced information.



Tracking due dates and overdue fines is difficult without automation.



A computerized system is required to manage library information efficiently.

Objectives

1

To maintain accurate records of books and library members.

2

To provide quick searching and checking of book availability.

3

To automate book issue and return operations.

4

To maintain transaction and borrowing history.

5

To reduce paperwork and minimize human errors.

6

To improve the efficiency and reliability of library management.

Entities identified



Member / Student

Person who borrows books.



Book

Library resource available for borrowing.



Author

Person who writes a book.



Publisher

Organization that publishes a book.



Librarian

Person who manages library operations.



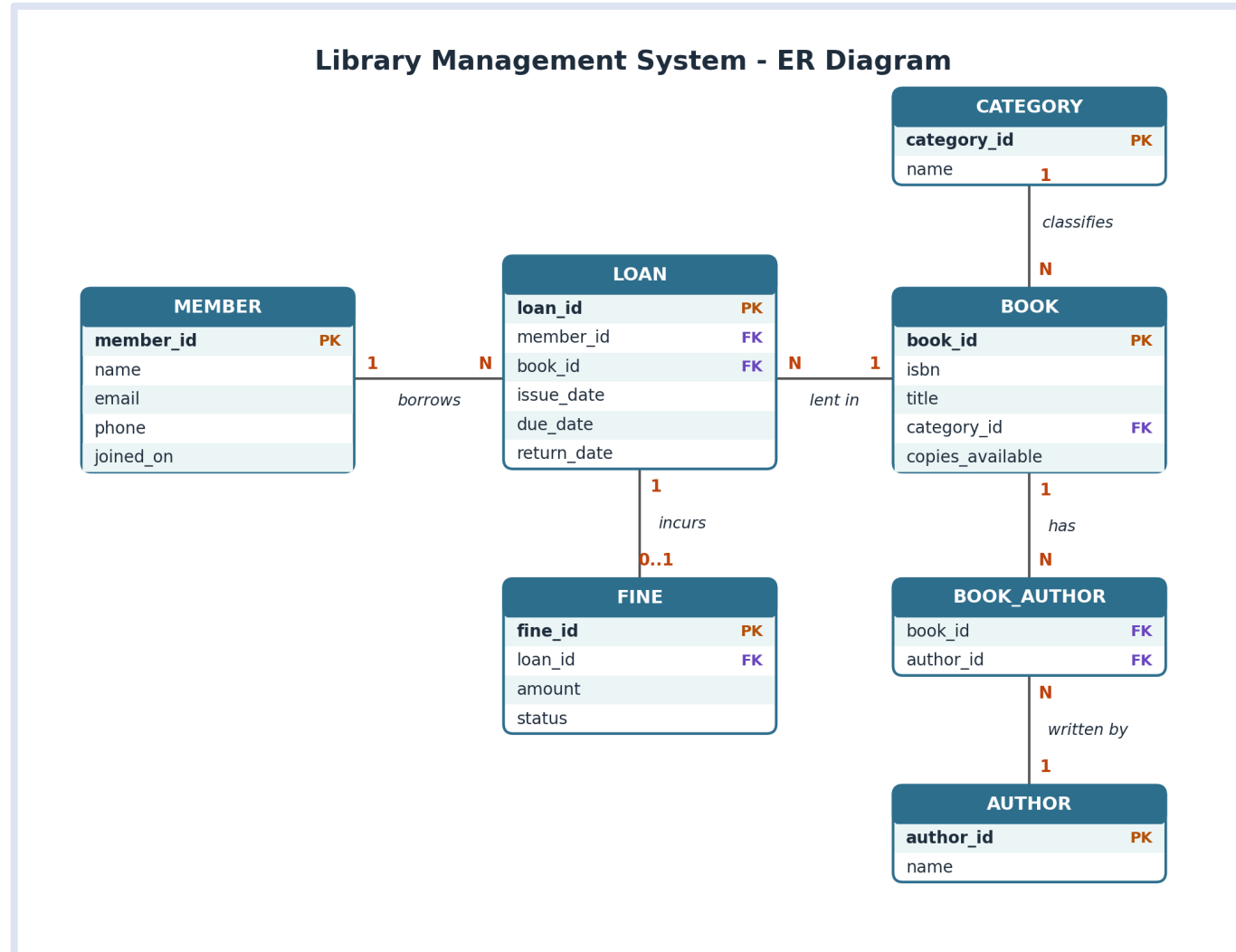
Issue / Return

Transaction related to borrowing and returning books.

Attributes

Entity	Attributes
Member	Member_ID, Name, Contact, Email, Address
Book	Book_ID, Title, ISBN, Category, Price, Availability
Author	Author_ID, Author_Name
Publisher	Publisher_ID, Publisher_Name, Contact
Librarian	Librarian_ID, Name, Contact, Email
Issue / Return	Transaction_ID, Book_ID, Member_ID, Issue_Date, Due_Date, Return_Date, Fine

ER diagram



Relationships & cardinality

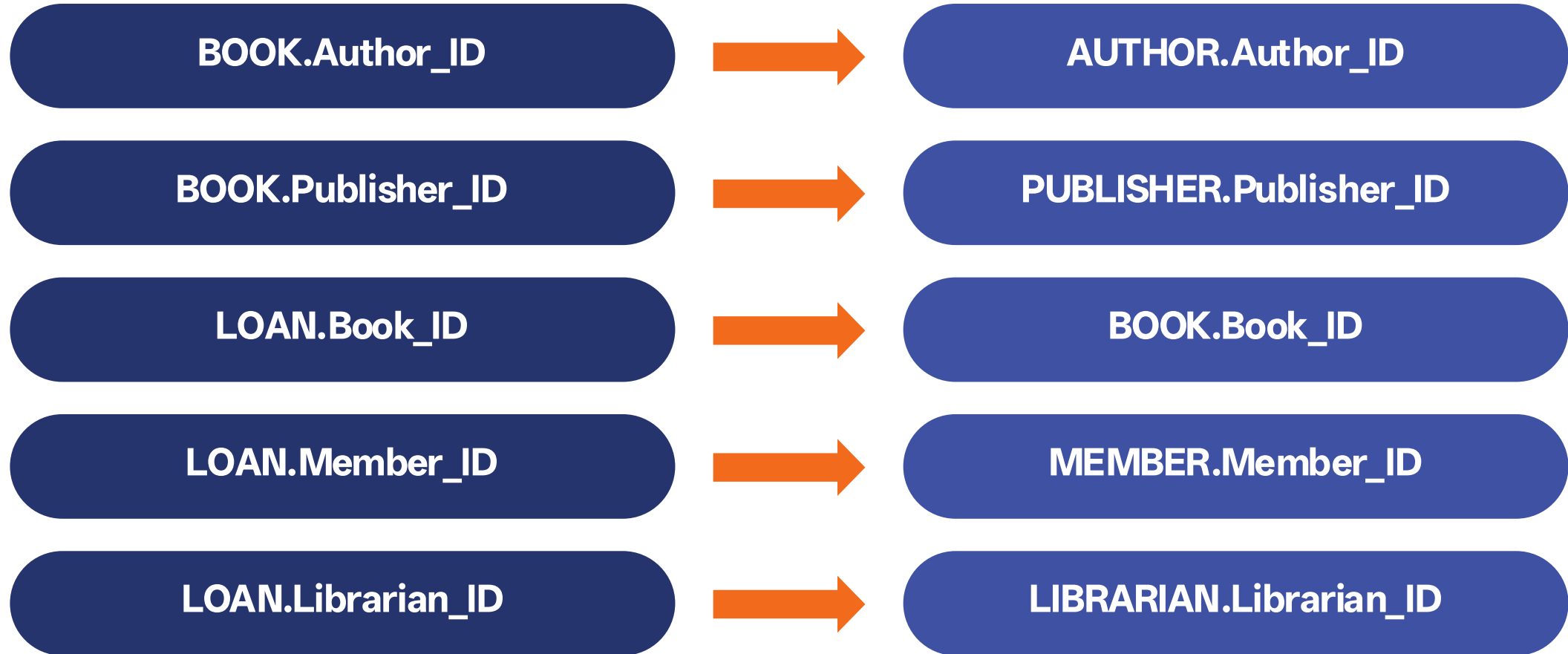
All major relationships are 1:N (one-to-many)



Primary & foreign keys

Table	Primary key (PK)	Foreign key (FK)
BOOK	Book_ID	Author_ID, Publisher_ID
MEMBER	Member_ID	None
AUTHOR	Author_ID	None
PUBLISHER	Publisher_ID	None
LIBRARIAN	Librarian_ID	None
LOAN	Loan_ID	Book_ID, Member_ID, Librarian_ID

Foreign key mappings



Relational database schema

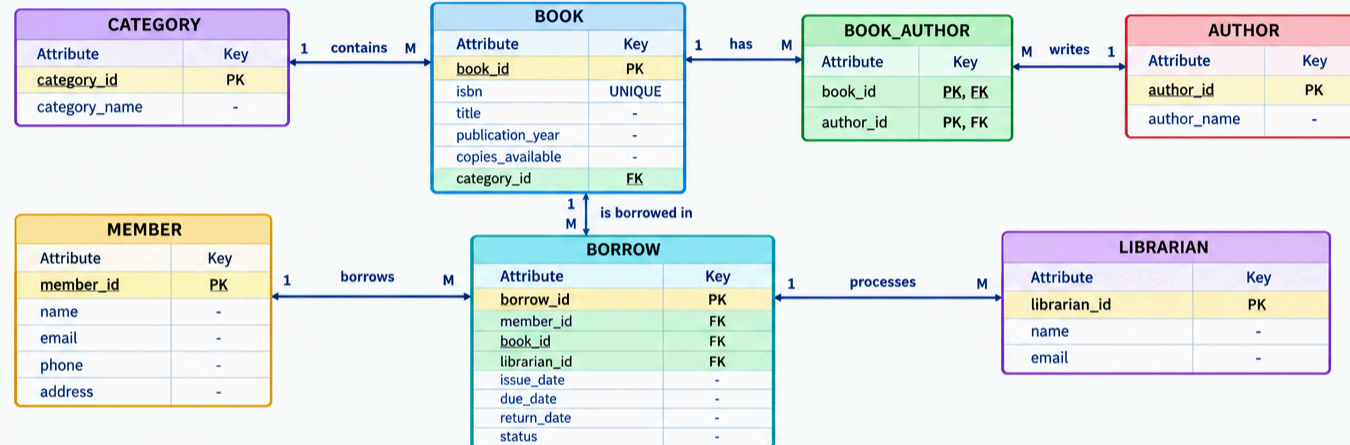


Library Management System

Review 1 – Database Design



6 Relational Database Schema



Relationships

No.	Relationships	Entities Involved	Cardinality	Description
1	CATEGORY – BOOK	CATEGORY ↔ BOOK	1 : M	One category can contain many books.
2	BOOK – BOOK_AUTHOR	BOOK ↔ BOOK_AUTHOR	1 : M	One book can have multiple authors (via bridge table).
3	AUTHOR – BOOK_AUTHOR	AUTHOR ↔ BOOK_AUTHOR	1 : M	One author can write multiple books (via bridge table).
4	BOOK_AUTHOR – BOOK	BOOK_AUTHOR ↔ BOOK	M : 1	A book can have multiple authors and an author can write multiple books (resolved by BOOK_AUTHOR).
5	MEMBER – BORROW	MEMBER ↔ BORROW	1 : M	One member can borrow many books.
6	BOOK – BORROW	BOOK ↔ BORROW	1 : M	One book can be borrowed many times.
7	LIBRARIAN – BORROW	LIBRARIAN ↔ BORROW	1 : M	One librarian can process many borrow transactions.

Legend

- Entity = Entity
- ◊ = Relationship
- Attribute = Attribute
- Primary Key (PK)
- Foreign Key (FK)
- Unique Constraint
- 1, M** = Cardinality

Database design summary

6

Entities

5

Binary relationships

6

Relational tables

6

Primary keys

5

FK mapping groups

1:N

Cardinality (all relationships)

Integrity constraints

PRIMARY KEY

FOREIGN KEY

NOT NULL

UNIQUE

CHECK

Project status



Review-I

Completed

- Entity identification
- Attributes
- ER model
- Relationships
- Keys
- Relational schema



Review-II

Next

- Implementation
- SQL queries / CRUD operations
- Testing
- Further validation

Conclusion

The Library Management System provides a structured relational design for managing books, members, authors, publishers, librarians and loan transactions. The proposed schema reduces redundancy, improves search and availability tracking, and enforces data integrity through database constraints.

NEXT STEP

Review-II will focus on database implementation, SQL operations, testing and validation.

Thank you

Library Management System



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